USED CAR PRICE ANALYSIS USING PYTHON

Project Report

Submitted by: Nithish kumar S
Department: B.Tech Artificial Intelligence and Data Science
College: Global Institute of Engineering and Technology

---

1. INTRODUCTION

The used-car market includes vehicles of different manufacturers, models, ages, fuel types, and transmission systems. Determining a reasonable price for a used car can be difficult because many factors may influence its value.

This project, titled **"Used Car Price Analysis Using Python,"** uses data analysis techniques to explore used-car prices and identify relationships between price and selected vehicle characteristics.

Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn are used to load, process, analyze, and visualize the dataset. Graphs help present the findings in an understandable way.

The project provides practical experience in data analysis and demonstrates how Python can be used to explore a real-world business problem.

2. OBJECTIVES

The main objectives of this project are:

1. To load and examine a used-car dataset using Python.
2. To understand the structure and quality of the data.
3. To identify missing values and duplicate records.
4. To compare average car prices across manufacturers.
5. To analyze prices by fuel type and transmission type.
6. To examine the relationship between vehicle age and price.
7. To examine the relationship between kilometres driven and price.
8. To present the findings through charts and visualizations.

9. To develop practical skills in Python-based data analysis.

3. DATASET DESCRIPTION

The project uses the CSV dataset named `car details v4.csv`.

The dataset contains approximately **2,059 records and 20 columns** describing used cars.

Important columns include:

Column	Description
Make	Manufacturer of the car
Model	Model of the car
Price	Listed price of the car
Year	Year of manufacture
Kilometer	Distance travelled
Fuel Type	Fuel used by the vehicle
Transmission	Manual or automatic transmission
Location	Location associated with the listing
Owner	Ownership information
Seller Type	Type of seller
Engine	Engine specification
Max Power	Maximum engine power
Drivetrain	Type of drivetrain

The dataset also contains technical specifications such as vehicle dimensions, seating capacity, and fuel tank capacity.

Note: The dataset's original source and price currency should be verified before assigning a specific currency to the price values.

4. TOOLS AND TECHNOLOGIES

The following tools and libraries were used:

- **Python:** Programming language used for the analysis.
- **Google Colab:** Cloud-based environment for running Python code.
- **Pandas:** Used to load, organize, and analyze tabular data.
- **NumPy:** Used for numerical operations.
- **Matplotlib:** Used to create charts and graphs.
- **Seaborn:** Used to create statistical visualizations.

5. METHODOLOGY

The project follows a step-by-step data analysis process.

5.1 Data Loading

The dataset was imported into Python using Pandas. The data was stored in a DataFrame so that its rows and columns could be examined.

5.2 Data Inspection

The dataset was inspected to understand its dimensions, column names, data types, and sample records.

This step helped identify the fields required for the analysis.

5.3 Data Quality Checking

The dataset was checked for missing values and duplicate records.

Missing values were present in some technical specification columns, including engine, maximum power, drivetrain, and vehicle dimensions. No missing values were reported in the core Make, Model, Price, Year, and Kilometer columns. No exact duplicate records were identified.

5.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand patterns in the dataset.

The analysis compared car prices across manufacturers, fuel types, and transmission types. It also explored how price varied with vehicle age and kilometres driven.

5.5 Data Visualization

Charts were created using Matplotlib and Seaborn.

The visualizations made it easier to compare groups, observe patterns, and communicate the findings.

5.6 Interpretation

The results were interpreted to identify associations between vehicle characteristics and listed prices.

These associations do not establish that a particular characteristic directly causes a price change.

6. RESULTS AND DISCUSSION

6.1 Price by Manufacturer

The average listed price was compared across car manufacturers.

This comparison helps identify differences in average prices between manufacturers. However, the averages may also reflect differences in vehicle models, age, mileage, and specifications.

6.2 Price by Fuel Type

Prices were compared across available fuel types.

This analysis helps explore whether vehicles using different fuel types have different average listed prices. Results for fuel types represented by very few records should be interpreted cautiously.

6.3 Automatic and Manual Transmission

The average listed prices of automatic and manual cars were compared.

The comparison provides an overview of price differences between transmission groups. It does not isolate transmission type from other factors, such as manufacturer, model, age, or engine specifications.

6.4 Vehicle Age and Price

The relationship between vehicle age and price was investigated.

The calculated correlation between vehicle age and price was approximately **−0.311**, indicating a moderate negative association in this dataset.

In general, this result suggests that older vehicles tended to have lower listed prices. However, individual vehicles may differ because of their condition, model, maintenance history, and other characteristics.

6.5 Kilometres Driven and Price

The relationship between kilometres driven and price was also examined.

The calculated correlation between kilometres driven and price was approximately **−0.244**, indicating a weak negative association in this dataset.

This suggests that vehicles with higher recorded kilometres tended to have lower listed prices, although the relationship was not strong.

6.6 Summary of Findings

The analysis produced the following main observations:

- Average listed prices differed across manufacturers.
- Price patterns varied across fuel types.
- Automatic and manual cars showed differences in their average listed prices.
- Vehicle age had a negative correlation with price of approximately −0.311.
- Kilometres driven had a negative correlation with price of approximately −0.244.

The results describe associations within the available dataset. They should not be treated as universal rules for every used car.

7. LIMITATIONS

This project has several limitations:

1. The analysis uses an existing dataset rather than a newly collected market survey.
2. The original data source and currency require verification.
3. Missing technical specifications may limit some comparisons.
4. Some fuel types may have too few records for reliable group comparisons.
5. Factors such as vehicle condition, accident history, service records, and local market demand may not be fully represented.
6. Correlation does not prove causation.
7. The project explores price patterns; it does not establish the accuracy of a predictive pricing model.

8. CONCLUSION

This project demonstrated how Python can be used to analyze used-car data and explore factors associated with listed prices.

Using Pandas, NumPy, Matplotlib, and Seaborn, the dataset was inspected, its quality was examined, and several comparisons and visualizations were created.

The results showed negative associations between vehicle age and price and between kilometres driven and price. Price differences were also explored across manufacturers, fuel types, and transmission types.

The project provided practical experience in data loading, data inspection, exploratory data analysis, visualization, and interpretation.

In conclusion, Python-based data analysis is a useful approach for exploring used-car market data. A more accurate pricing system would require additional relevant features, verified data sources, and further statistical or machine-learning evaluation.

9. FUTURE ENHANCEMENTS

The project could be improved by:

- Collecting more recent used-car listings.
- Verifying the dataset's original source and currency.
- Including vehicle condition and service history where available.
- Investigating outliers and unusual price values.
- Comparing additional vehicle characteristics.
- Developing and evaluating a machine-learning model to estimate used-car prices.
- Testing any predictive model on data not used during training.

10. REFERENCES

1. Pandas Documentation — https://pandas.pydata.org/docs/
2. NumPy Documentation — https://numpy.org/doc/
3. Matplotlib Documentation — https://matplotlib.org/stable/
4. Seaborn Documentation — https://seaborn.pydata.org/